PMF (Probability Mass Function)

PMF is referred to as the probability of discrete random variable which is equal to a particular value. It is represented as f(x) = P (X = x)

where

- X is discrete random variable
- x is the specified value.

Example:
Let is dice is rolled then the probability of getting a number equal to 4 is an example of probability mass function. The sample space for the given event is {1, 2, 3, 4, 5, 6} and X be the random variable for getting a 4. The probability mass function evaluated for X = 4 is 1/6.

Probability Mass Function Formula in Binomial Distribution

Binomial distribution with the number of outcomes, probability of success and probability of failures. The PMF formula in Binomial distribution is given by:

$$P(X = x) = {}^{n}C_{x}\, p^{x}\, (1 - p)^{n-x}$$

where,

- n is Number of Outcomes
- p is Probability of Success
- (1-p) is Probability of Failure

Probability Mass Function Formula in Poisson Distribution

Poisson distribution deals with the mean and the number of independent events occurred in specific interval of time. The formula for the probability mass function in Poisson distribution is given by:

$$P(X = x) = [\lambda^{x} e^{-\lambda}] / x!$$

where,

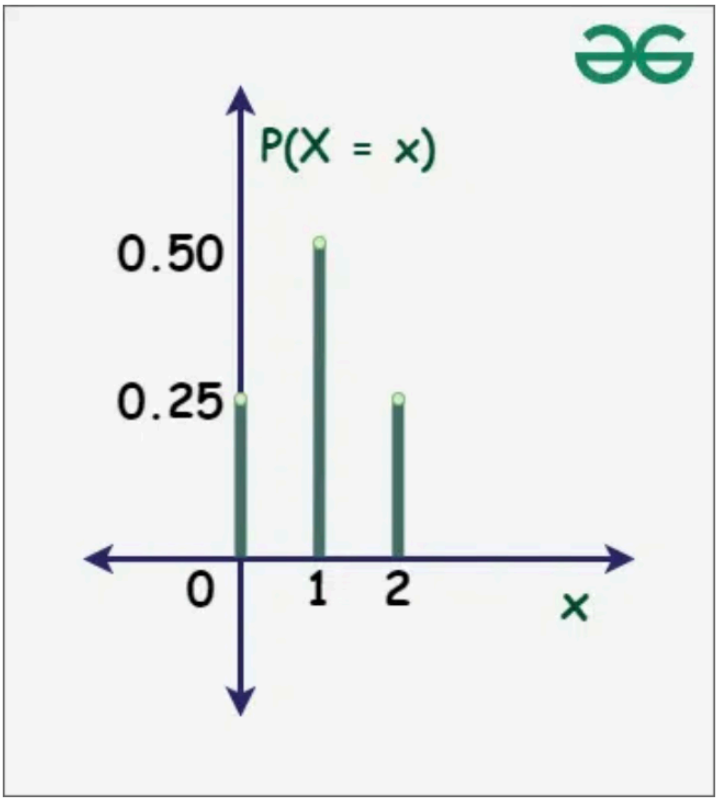
- λ is Mean

Probability Mass Function Table and Graph

Table that represents the probability mass function with the value of the random variables is called the probability mass function. Let a coin is tossed two times and X be the random variable representing the number of tails then, the probability mass function table for above event is given below.

x	0{HH}	1{HT, TH}	2{TT}
P(X = x)	1/4	2/4 = 1/2	1/4

Probability mass function graph for above table is given below:



Probability Mass Function

Properties of Probability Mass Function

Some properties of the probability mass function are listed below.

- **$f(x) = P(X = x) > 0$** hence, probability is always positive.
- **$\sum_{x \in S} f(x) = 1$** hence, sum of all probabilities equals to 1.
- **$P(X \in E) = \sum_{x \in E} f(x)$** hence, probability of event E is given by sum of probabilities of values of x in E. It is used to determine CDF.

PROBABILITY MASS FUNCTION	PROBABILITY DENSITY FUNCTION
The PMF is the probability that a discrete random variable takes at an exact value.	The PDF is the probability that a continuous random variable takes at a specified interval.
The PMF deals with the discrete random variables.	The PDF deals with the continuous random variables.
PMF is evaluated at specific point.	PDF is evaluated at specified interval
f(x) = P (X = x)	P(x) = F'(x) where, F(x) is CDF

Uses of Probability Mass Function

Probability Mass Function (PMF) is a fundamental concept in probability theory and statistics, particularly when dealing with discrete random variables. Some uses of the PMF are:

- **Describing Discrete Distributions:** PMF is used to define the distribution of discrete random variables such as the number of heads in a series of coin flips, the number of customers arriving at a store in a given hour, or the number of defective items in a batch.
- ***Calculating Probabilities:** PMF allows for the calculation of the probability that a random variable takes on a specific value.
- **Modeling and Simulations:** In simulation studies and probabilistic modeling, PMFs are used to generate random samples and to simulate scenarios involving discrete random variables.
- **Information Theory:*** PMFs play a crucial role in information theory, particularly in the calculation of entropy, which measures the uncertainty or randomness of a random variable.

***Example 1:** Probability mass function is given by: f(x) = ax2 for x = 0, 1, 2 then, find the value of a.

***Solution:**

To find the value of a we use the PMF property.

$\sum x\in Sf(x) = 1$

f(0) + f(1) + f(2) = 1
a × 02 + a × 12 + a × 22 = 1>
a × (02 + 12 + 22) = 1
a × (0 + 1 + 4) = 1
5a = 1
a = 1/5

Example 2: From the below probability mass function table determine CDF.

X	1	2	3	4
P(X = x)	0.2	0.05	0.25	0.5

Solution:

To find the CDF for P(X < 3) we use the property of PMF.
P (X ∈ E) = ∑x∈ Ef(x)
P(X < 3) = P(X = 1) + P(X = 2)
P(X < 3) = 0.2 + 0.05
P(X < 3) = 0.25

So, CDF of P(X < 3) = 0.25

***Example 3:** Table below represents the PMF for a random variable X. Find the value of p.

X	2	4	6
P(X = x)	p	p + 0.5	0.2

Solution:

To find the value of p we use formula.
∑****x∈ S***f(x) = 1
p + p + 0.5 + 0.2 = 1
2p + 0.7 = 1
2p = 1 - 0.7
2p = 0.3
p = 0.3 / 2
p = 0.15

***Example 4:** Find probability of good number of products = 8 if there 10 products and the probability of good product is 0.95.

Solution:

We will solve above question using Binomial distribution.

Here, n = 10, x = 8, p = 0.95

PMF of Binomial distribution is given by:

$$P(X = x) = {}^{n}C_{x}\, p^{x}\, (1 - p)^{n-x}$$

$$P(X = 8) = {}^{10}C_{8}\, (0.95)^{8}\, (1 - 0.95)^{10 - 8}$$

$$P(X = 8) = 45 \times 0.663 \times (0.05)^{2}$$

$$P(X = 8) = 29.835 \times 0.0025$$

$$P(X = 8) = 0.0746$$

Probability of 8 good products = 0.0746

Example 5: There are 10 pens, and the probability of defective pens is 0.1 then find the probability of 2 defective pens.

Solution:*

Mean $= \lambda = np = 10 \times 0.1 = 1$

$P(X = x) = [\lambda^{x} e^{-\lambda}] / x!$

$P(X = 2) = [1^{2} \times e^{-1}] / 2!$

$P(X = 2) = (0.3679) / 2$

$P(X = 2) = 0.184$

Probability of 2 defective pens is 0.184